

3D Engines in Practice (CEN 4930)

Summer 2026

M4, T2, W4

Instructor

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Teaching Assistants

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Course Description

Focus on technical knowledge & skill in design, conceptualization, & visualization for three-dimensional engine content production. Incorporates scene / agent development, multi-disciplinary teams & pipelines, and production process. Project-based, culminating in a practical 2D/2.5D/3D application. (3)

Course Pre-Requisites / Co-Requisites

Some prior programming experience

Course Objectives

By the end of the semester, successful students should be able to:

- Conceptualize and plan the systems, design, and visual aesthetics for a 2D/2.5D application.
- Develop an industry standard design document.
- Consider and integrate designs regarding usability and user experience.
- Effectively implement core principles of the design production process.
- Create a final application build including multiple levels in a cohesive executable package.

Relation to Program Outcomes (ABET):

Outcome	Coverage*
1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	Medium
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	
3. An ability to communicate effectively with a range of audiences	
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	Low
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	Medium
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	Medium

*Coverage is given as high / medium / low. Empty box indicates this outcome is not covered / assessed.

Required Textbooks and Software

There are no required materials for this course. All materials will be provided by the instructor. The College of Engineering requires students to have a mobile computing device (laptop).

Course Schedule

Wk	Subject	Quiz	Course Work	Milestones
1	Syllabus	Q0		-
2	Game / Sim History & Design	Q1, Q2	Ex0 (Design)	Pitch, Teams
3	Setup, Engine Basics, & Sim Loop	Q3, Q4	Ex1 (PewPew)	Design Draft
4	Development & Version Control	Q5	Ex2 (Git)	Peer Evaluation
5	Signals & Design Patterns	Q6, Q7	Ex3 (Input)	Check-In, Pre-Alpha
6	Physics & Collisions	Q8, Q9	Ex4 (Collisions)	Peer Evaluation
7	Animation	Q10	-	Check-In, Prototype
8	Prototype Presentations			Presentation Reviews
9	User Experience & User Interfaces	Q11, Q12	-	Peer Evaluation
10	ASYNCHRONOUS WORK WEEK			
11	Lighting & Audio	Q13	-	Lighting, Audio
12	Irrationality; Inventory Factories	Q14, Q15	-	Peer Evaluation
13	Post-Mortem Presentations			Check-In, Release, Peer Eval.

Attendance & Make-Up Policy

Requirements for attendance and make-ups for exams and other work in this course are consistent with university policies: <https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/>

Evaluation of Grades

Assignment	Undergrad	Graduate
Project Milestones		
Pitch	50 (5%)	50 (5%)
Final Design	50 (5%)	50 (5%)
Prototype Presentation	80 (8%)	80 (8%)
Post-Mortem (Pres.)	140 (14%)	140 (14%)
Prototype Milestone	100 (10%)	100 (10%)
Production Release	200 (20%)	200 (20%)
Interaction		
Stakeholder Mtgs. (5-10)	50 (5%)	30 (3%)
Pres. Reviews (4)	10 x 4 (4%)	5 x 4 (2%)
Peer Evaluations (4)	10 x 4 (4%)	5 x 4 (2%)
Professionalism	10 (1%)	
General Coursework		
Exercises (5)	20 x 5 (10%)	14 x 5 (7%)
Quizzes (16-drop-2)*	10x14 (14%)	10x14 (14%)
Topic Pres. (Grad)	-	100 (10%)

Grading Policy

Percent	Grade	Pts	Percent	Grade	Pts
93 – 100	A	4.00	73 - 76	C	2.00
90 – 92	A-	3.67	70 – 72	C-	1.67
87 – 89	B+	3.33	67 - 69	D+	1.33
83 – 86	B	3.00	63 - 66	D	1.00
80 – 82	B-	2.67	60 – 62	D-	0.67
77 – 79	C+	2.33	00 – 59	E	0.00

Graduate-Level Work

Students in the graduate section of the course are expected not only to play a leadership role in their teams but also to act as mentors and domain experts who will guide younger practitioners. Graduate students are expected to spend approximately 133% of the time on the course compared to undergraduate students. This will include a topical presentation for 8% of the final grade in the class.

Percentages will be rounded to the nearest whole percentage for the purposes of assigning letter grades. For more information on UF grading policy, see [the catalog](#).

Exercises. Short assignments intended to reinforce fundamental engine concepts.

Quizzes. Taken each day on the content previously assigned content for reinforcement.

Evaluations, Reviews, and Meetings. Honest, constructive meetings, reviews, & evaluations of peers.

Project Milestones. Student teams will complete the following milestones throughout this course:

Project Pitch – One-minute “elevator pitch” for project

Design Draft – A preliminary / draft plan for the project

Final Design – Shows key elements (hardware sketch, back-end dataflow chart, and/or UI wireframe)

Design Prototype – “Proof of Life”; demonstration of “vertical slice” of functionality

Prototype Presentation – Live presentation of prototype work to peer audience

Production Release – Feature complete version of the project (final submission)

Post-Mortem Presentation – Post-project evaluation of project challenges, successes, & overall outcomes

Student Teams

Most software requires many disparate skill-sets. Few students will have all these skills; instead, student should form groups with the goal of create well-rounded teams. All students will be responsible for completing peer reviews that, on their honor, are correct and accurate. Students will also be required to utilize GitHub or another source control system that may be used to task review and consideration. Students are also required to track features and issues using a professional tool (e.g., Trello and/or GitHub issues).

Based on peer reviews, instructors will intervene to correct behavior as necessary to maintain well-functioning teams. All team members are expected to contribute to the project on equal terms; as such, grade adjustments may be made if it is determined by faculty that student(s) are not acting productively as part of the team.

Finally, students are expected to be available during regular business hours (JST) to contribute to the team’s efforts, meetings, and necessary communications. Teams are expected to meet **synchronously**, at a minimum, on a biweekly basis, but more frequent meetings are recommended for the health of the projects. Meetings are not required to be formal, but to build community, communication channels, and rapport; for example, some students may choose to engage in a short formal meeting each week but meet informally for co-working on other days.

Code & Schematic Submissions

Functionality is key to success in computing, so it is **extremely important** that the guidelines are followed. Failure to follow these instructions will result in penalties.

- Code must compile / run in debug and release mode, and schematics must be error-free. Debug information should never be released in the final version of a software project. **Projects that do not compile AND run will be marked zero.**
- Include only those files specified by the documents in your archives. Projects should have only directory structure explicitly mentioned in the documentation (i.e., relevant files and folders should be submitted in the root of the zip file.) It should be possible to open the archive, copy your files directly into the project, compile, and then run the project without further steps. Packaging errors will result in a grade of **zero**.

Course Expectations

Read and adhere to the syllabus. Emails requesting information contained in the syllabus will receive the lowest priority for response with no guaranteed turnaround. Practically, this means responses will come only after the remaining email queue of the instructor is otherwise empty. This condition occurs approx. once every 24 months.

Students should not distract others in class. Students should avoid watching videos; playing games; talking; sleeping; howling; biting toenails; screeching like a banshee; & other distracting behaviors.

Students must act with honor; academic dishonesty will be strictly addressed. Sharing / copying, “borrowing” of code structure, discussing code structure, looking at code from another student, providing such code, and plagiarism, in addition to other dishonest behaviors, are considered academic dishonesty. No information regarding assignment solutions may be shared by students except at a conceptual level. If students implement algorithms from other sources, they must be cited. Students may not copy code from the Internet or other sources under any circumstances. Any student found to have violated these rules, whether a provider or receiver or unauthorized help, will be assigned a grade of E (failing) in the course and referred to the Honor Court. **When in doubt, ask.**

All assignments are due by the time listed on Canvas. Projects and homework with a cascading deduction: one (1) weekday late for 10% penalty; two (2) for 25% penalty; or three (3) for 50% penalty. Quizzes and presentations may not be completed late for credit except with instructor approval for extenuating circumstances (see below).

Quiz, presentation, & meeting make-ups are permitted only in extenuating circumstances. For make-up consideration students will be required to submit written documentation from a reputable source as evidence. For any planned event (e.g., a wedding), the student is expected to contact the instructor no less than two weeks in advance for consideration. Please note that there is no guarantee that requests will be accommodated. Social, networking, and club events may be taken into consideration strictly at the discretion of the instructor.

Grade reviews must be requested within one week of a grade being posted. After two weeks, no grades will be revisited. In the event of a grade review, the entire assignment will be reviewed.

Students should visit office hours for project help and grade questions. Do not send email to, send private messages to, or “@” instructors or TAs about project help or grades. The TAs and instructor will often try to answer questions in the chat when possible, but the way to get personalized help is to visit or make arrangements!

Important non-project correspondence be via email. The chat system is helpful for simple questions and allows students to help one another, but students should not expect responses to important questions via chat. Please allow 48 business hours for responses; instructors and TAs have many responsibilities and respond as is practical.

Student Privacy

There are federal laws protecting your privacy with regards to grades earned in courses and on individual assignments. For more information, please see: <https://registrar.ufl.edu/ferpa.html>

Students Requiring Accommodations

Students with disabilities requesting accommodations should first register with the Disability Resource Center (352-392-8565, <https://www.dso.ufl.edu/drc>) by providing appropriate documentation. Once registered, students will receive an accommodation letter which must be presented to the instructor when requesting accommodation. Students with disabilities should follow this procedure as early as possible in the semester.

Course Evaluations

Students are expected to provide feedback on the quality of instruction in this course by completing online evaluations at <https://evaluations.ufl.edu/evals>. Evaluations are typically open during the last two or three weeks of the semester, but students will be given specific times when they are open. Summary results of these assessments are available to students at <https://evaluations.ufl.edu/results/>.

University Honesty Policy

UF students are bound by The Honor Pledge which states, “We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: “On my honor, I have neither given nor received unauthorized aid in doing this assignment.” The Honor Code (<https://sccr.dso.ufl.edu/policies/student-honor-code-student-conduct-code/>) specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. If you have any questions or concerns, please consult with the instructor or TAs in this class.

Commitment to a Safe and Inclusive Learning Environment

The Herbert Wertheim College of Engineering values broad diversity within our community and is committed to individual and group empowerment, inclusion, and the elimination of discrimination. It is expected that every person in this class will treat one another with dignity and respect regardless of gender, sexuality, disability, age, socioeconomic status, ethnicity, race, and culture.

If you feel like your performance in class is being impacted by discrimination or harassment of any kind, please contact your instructor or any of the following:

- Your academic advisor or Graduate Program Coordinator
- Director of Human Resources, 352-392-0903
- Associate Dean of Student Affairs, 352-392-2177
- Associate Dean of Academic Affairs, 352-392-0943

Software Use

All faculty, staff, and students of the University are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against University policies and rules, disciplinary action will be taken as appropriate. We, the members of the University of Florida community, pledge to uphold ourselves and our peers to the highest standards of honesty and integrity.

Campus Resources:

Health and Wellness

U Matter, We Care:

Your well-being is important to the University of Florida. The U Matter, We Care initiative is committed to creating a culture of care on our campus by encouraging members of our community to look out for one another and to reach out for help if a member of our community is in need. If you or a friend is in distress, please contact umatter@ufl.edu so that the U Matter, We Care Team can reach out to the student in distress. A nighttime and weekend crisis counselor is available by phone at 352-392-1575. The U Matter, We Care Team can help connect students to the many other helping resources available including, but not limited to, Victim Advocates, Housing staff, and the Counseling and Wellness Center. Please remember that asking for help is a sign of strength. In case of emergency, call 9-1-1.

Counseling and Wellness Center: <https://counseling.ufl.edu>, and 392-1575; and the University Police Department: 392-1111 or 9-1-1 for emergencies.

Sexual Discrimination, Harassment, Assault, or Violence

If you or a friend has been subjected to sexual discrimination, sexual harassment, sexual assault, or violence contact the [Office of Title IX Compliance](#), located at Yon Hall Room 427, 1908 Stadium Road, (352) 273-1094, title-ix@ufl.edu

Sexual Assault Recovery Services (SARS)

Student Health Care Center, 392-1161.

University Police Department at 392-1111 (or 9-1-1 for emergencies), or <http://www.police.ufl.edu/>.

Academic Resources

E-learning technical support, 352-392-4357 (select option 2) or e-mail to Learning-support@ufl.edu. <https://lss.at.ufl.edu/help.shtml>.

Career Connections Center, Reitz Union, 392-1601. Career assistance / counseling; <https://career.ufl.edu>.

Library Support, <http://cms.uflib.ufl.edu/ask>. Various ways to receive assistance with respect to using the libraries or finding resources.

Teaching Center, Broward Hall, 392-2010 or 392-6420. General study skills and tutoring. <https://teachingcenter.ufl.edu/>.

Writing Studio, 302 Tigert Hall, 846-1138. Help brainstorming, formatting, and writing papers. <https://writing.ufl.edu/writing-studio/>.

Student Complaints Campus: <https://sccr.dso.ufl.edu/policies/student-honor-code-student-conduct-code/>; <https://care.dso.ufl.edu>.

On-Line Students Complaints: <https://distance.ufl.edu/state-authorization-status/#student-complaint>.